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Muerte celular y cáncer

Sandra Arango Varela
Maestría Ingeniería Biomédica

Hacia una era de Universidad y Humanidad

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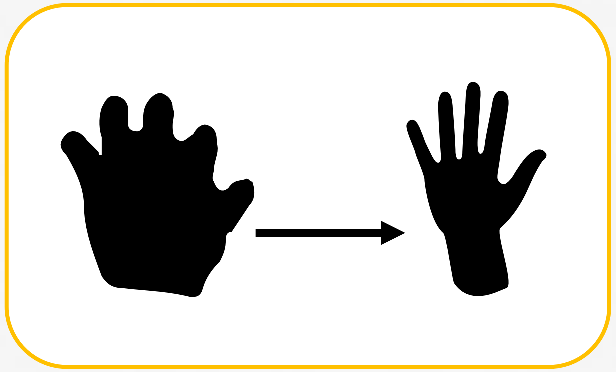
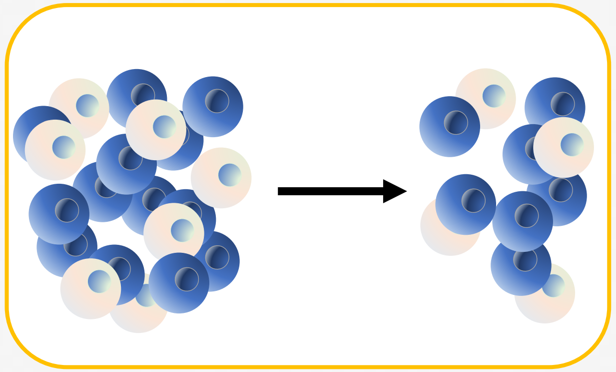
La muerte celular es una función biológica necesaria, involucrada en el crecimiento, la homeostasis y el mantenimiento de células sanas.

- La muerte celular programada desempeña un papel fundamental en el desarrollo animal y en la homeostasis de los tejidos.
- El mecanismo de la muerte celular está involucrado durante

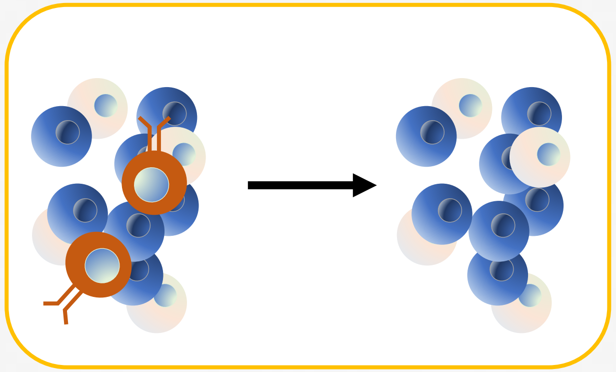
Arrangement of cells during morphogenesis



Regulation of cell number during homeostasis



Elimination of abnormal cells during pathogenesis



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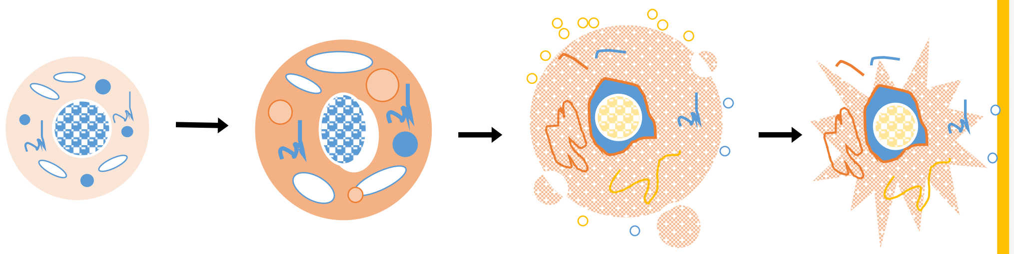
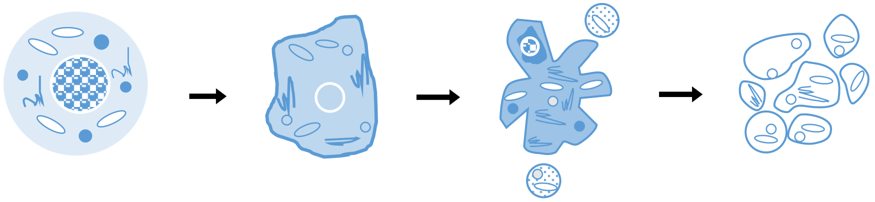
La muerte celular ocurre típicamente como resultado de necrosis o de una muerte celular programada, como la apoptosis

Necrosis occurs when cells die in response to injury or cellular stress, by **swelling and rupturing**



Apoptosis occurs when cells die during (e.g.) homeostasis, by condensation – without losing membrane integrity - and **removal by phagocytosis**



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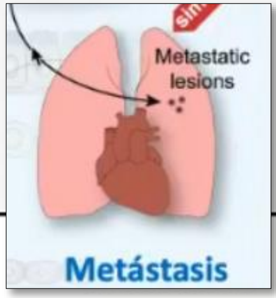
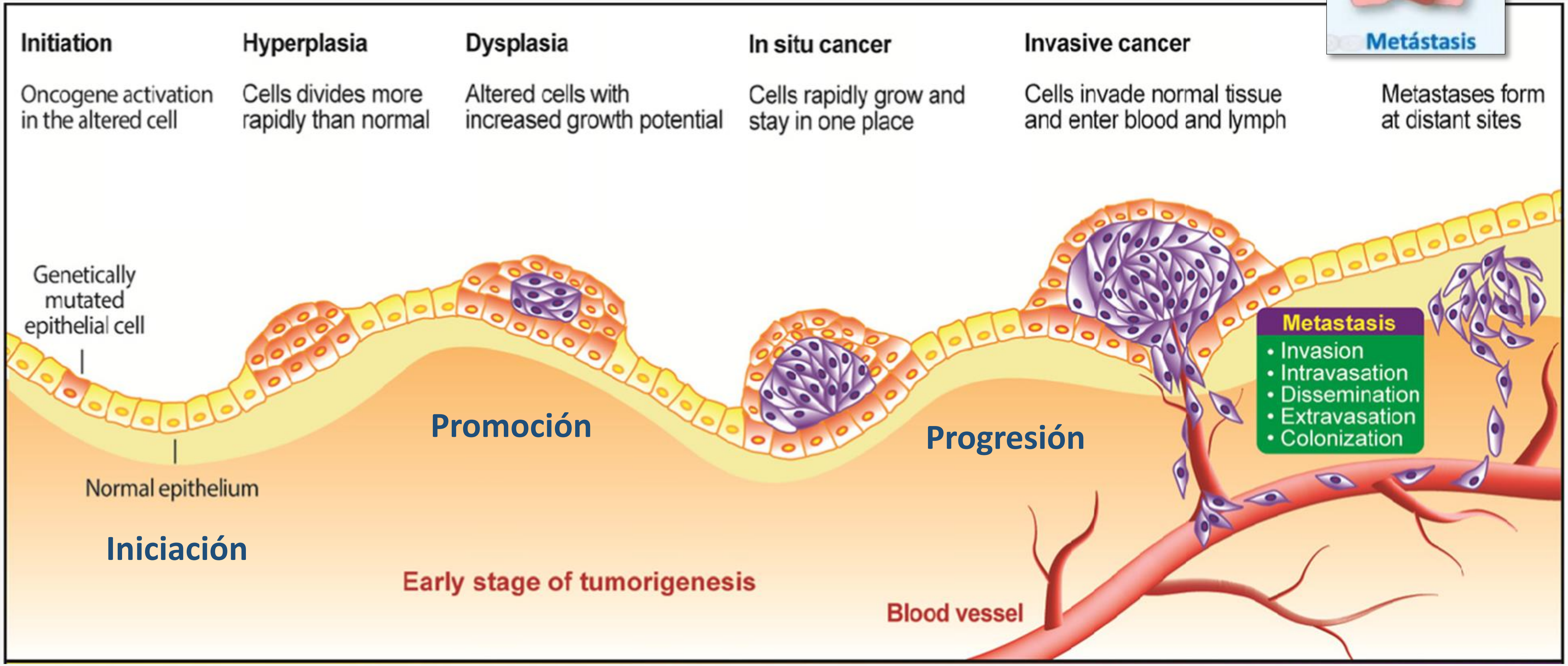
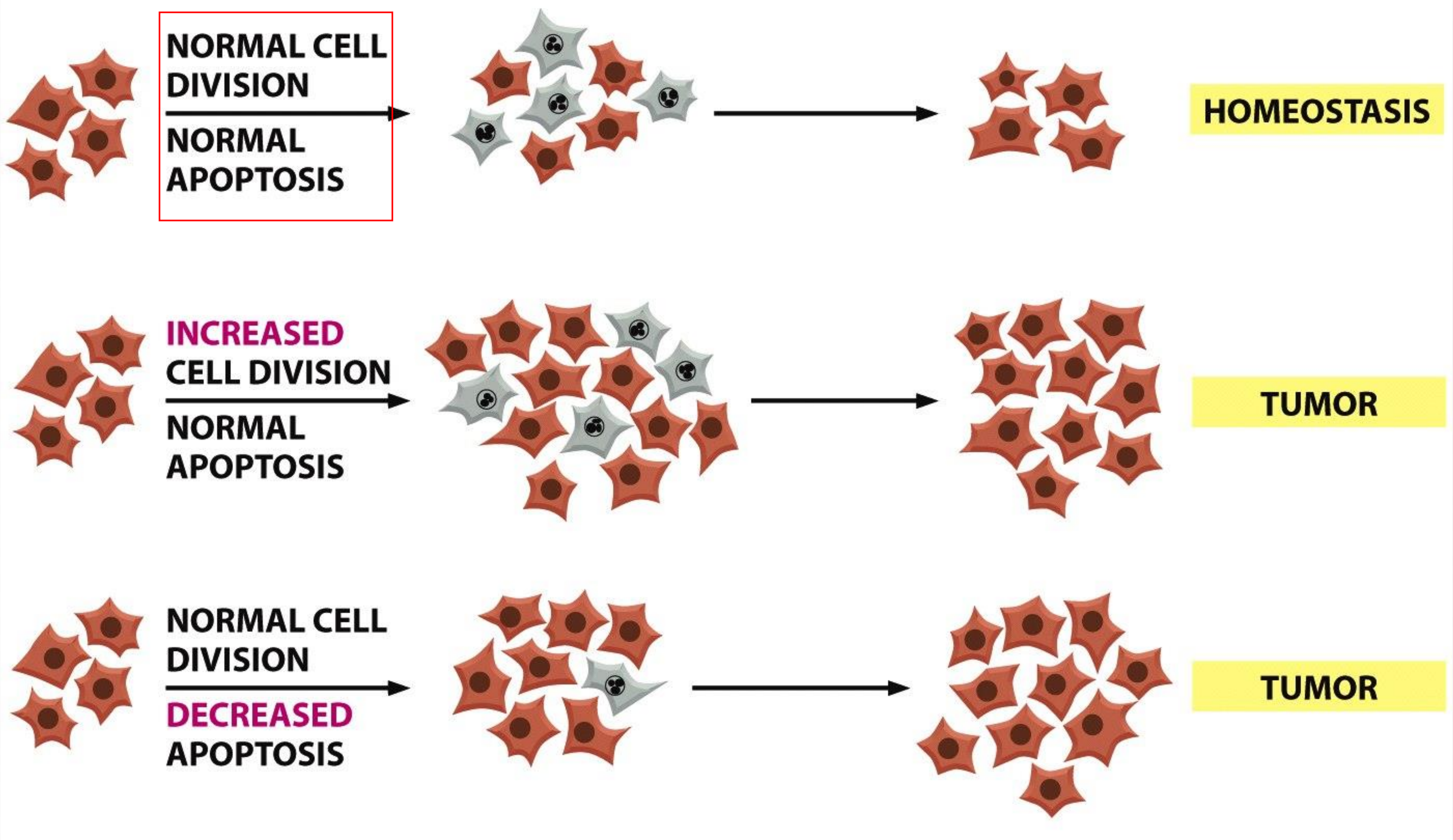
¿Qué es el cáncer?





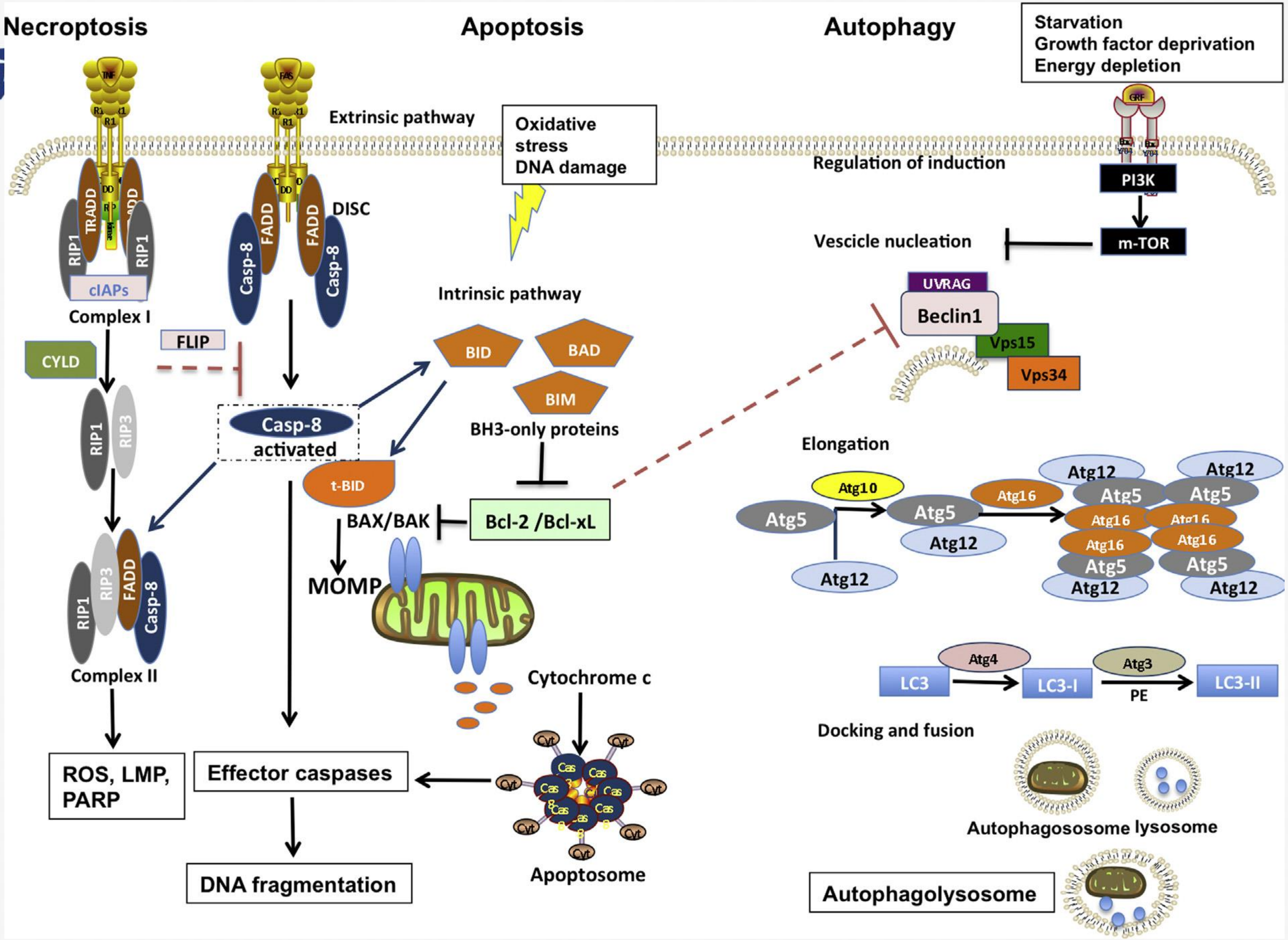
Li, X., He, S. & Ma, B. Autophagy and autophagy-related proteins in cancer. 2020

Cáncer = desbalance entre división celular y muerte

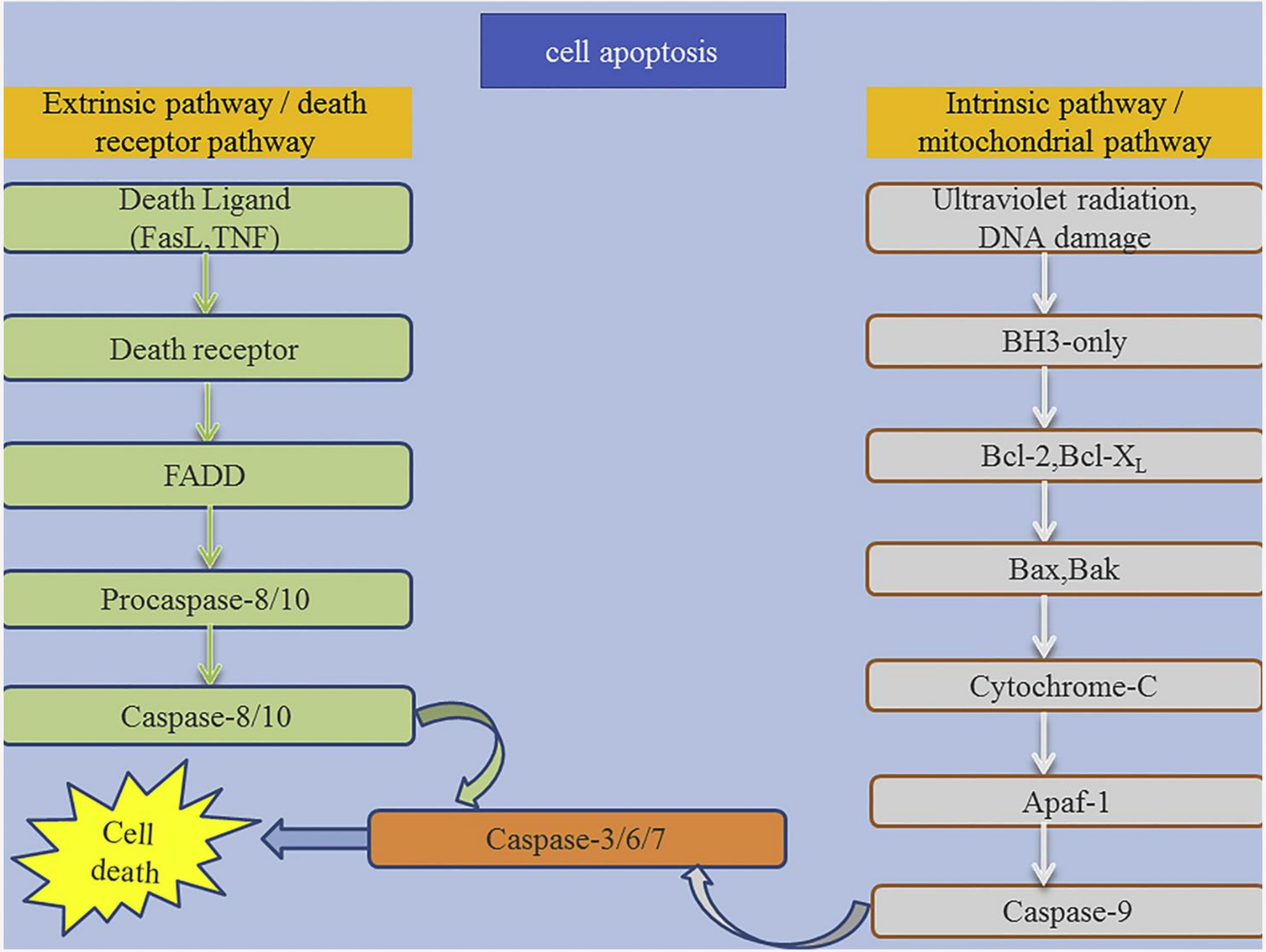


Cell death mode	Morphological features	Notes
Apoptosis	• Retracción de pseudópodos • Reducción del volumen celular y nuclear (picnosis) • Fragmentación nuclear (cariorrexis) • Modificación menor de los organelos citoplasmáticos • Formación de vesículas en la membrana plasmática (membrane blebbing) • Fagocitosis por células fagocíticas residentes, in vivo	'Apoptosis' is the original term introduced by Kerr *et al*.[14] to define a type of cell death with specific morphological features. Apoptosis is NOT a synonym of programmed cell death or caspase activation.
Autophagy	• Ausencia de condensación de la cromatina • Vacuolización masiva del citoplasma • Acumulación de vacuolas autofágicas (de doble membrana) • Escasa o nula captación por células fagocíticas, in vivo	'Autophagic cell death' defines cell death occurring with autophagy, though it may misleadingly suggest a form of death occurring by autophagy as this process often promotes cell survival.[15],[16]
Necrosis	• Tumefacción citoplasmática (oncosis) • Ruptura de la membrana plasmática • Tumefacción de los organelos citoplasmáticos • Condensación moderada de la cromatina	'Necrosis' identifies, in a negative fashion, cell death lacking the features of apoptosis or autophagy.[4] Note that necrosis can occur in a regulated fashion, involving a precise sequence of signals.

Cell Death Differ. 2009, 16(1): 3–11

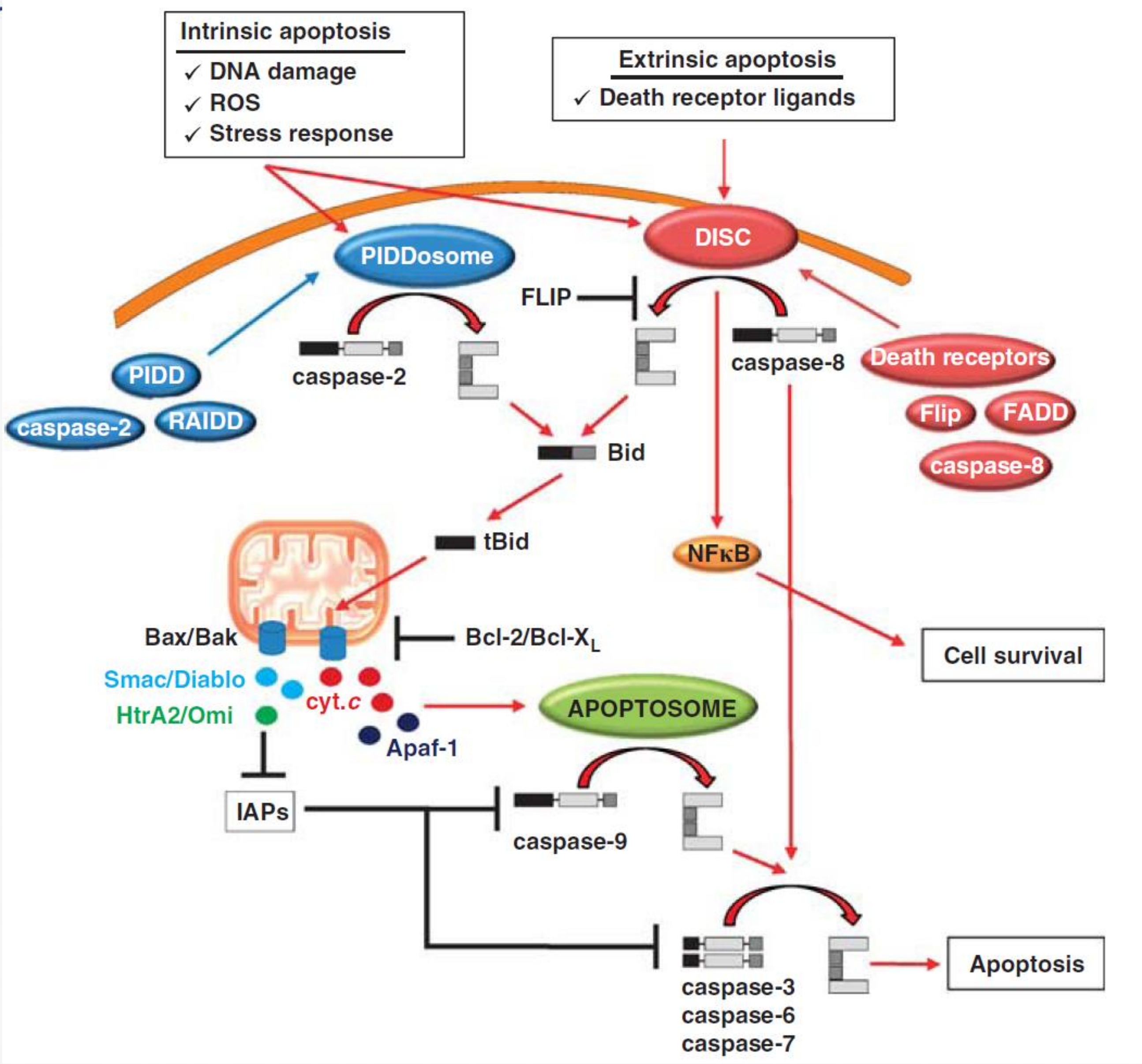


Necroptosis
Apoptosis
Autophagy
Starvation
Growth factor deprivation
Energy depletion
Extrinsic pathway
Oxidative stress
DNA damage
Regulation of induction
PI3K
m-TOR
Vescicle nucleation
DISC
TRADD
RIP1
cIAPs
Complex I
FADD
Casp-8
Intrinsic pathway
UVRAG
Beclin1
Vps15
Vps34
FLIP
CYLD
BID
BAD
BIM
BH3-only proteins
RIP1
RIP3
Casp-8
activated
Elongation
t-BID
BAX/BAK
Bcl-2 /Bcl-xL
MOMP
Atg10
Atg5
Atg16
Atg12
Complex II
Cytochrome c
Atg4
Atg3
LC3
LC3-I
LC3-II
PE
Docking and fusion
ROS, LMP, PARP
Effector caspases
Autophagosome
lysosome
DNA fragmentation
Apoptosome
Autophagolysosome



cell apoptosis
Extrinsic pathway / death receptor pathway
Death Ligand (FasL,TNF)
Death receptor
FADD
Procaspase-8/10
Caspase-8/10
Intrinsic pathway / mitochondrial pathway
Ultraviolet radiation, DNA damage
BH3-only
Bcl-2,Bcl-X_L
Bax,Bak
Cytochrome-C
Apaf-1
Caspase-9
Caspase-3/6/7
Cell death

Cascadas de Caspasas



La formación de complejos multicompontentes desencadena la **dimerización de las caspasas iniciadoras**, suficiente para su activación

•**DISC:** Complejo de señalización inductor de la muerte

•**Apoptosoma**

•**(PIDD)osoma:** Proteína inducida por **p53** con un dominio de muerte

Cell Death and Differentiation (2011) 18, 1441–1449

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Sustratos de Caspasas

Nuclear	Lamins, nucleoplasmin, the SR protein 70K U1, hnRNP C, RNA Pol I upstream binding factor, the p53 regulator MDM2, pRB, p27 ^{Kip} and p21^{Cip}
DNA related	MCM3, Repair enzymes including Rad51, poly-ADP-ribose polymerase (PARP), topoisomerase, inhibitor of caspase activated DNase, (iCAD/DFF45)
Cytoskeleton	actin, gelsolin, spectrin, keratin
Cytoplasmic	ß-catenin, Bcl-2
Protein kinases	DNA dependent protein kinase, protein kinase C, CAM kinase, focal adhesion kinase, MAP and ERK kinases, Raf1, Akt1/protein kinase B, ROCK I.

1. hnRNP C: Heterogeneous Nuclear Ribonucleoprotein C
2. RNA Pol I: RNA Polymerase I
3. MDM2: Mouse Double Minute 2 Homolog
4. pRB : Retinoblastoma Protein
5.p27 Kip / p21 Cip: Cyclin-dependent kinase inhibitor
6. MCM3: Minichromosome Maintenance Complex Component 3
7. Rad51: DNA Repair Protein Rad51
8. PARP: Poly (ADP-ribose) Polymerase
9. iCAD/DFF45: Inhibitor of Caspase-Activated DNase / DNA Fragmentation Factor 45 kDa subunit

10. Bcl-2 : B-cell Lymphoma 2
11. DNA-PK (DNA dependent protein kinase): DNA-dependent Protein Kinase
12. PKC (Protein kinase C)
13. CAM kinase / CaMK : Calmodulin-dependent Kinase
14. FAK (Focal adhesion kinase):
15. MAP kinases / ERK kinases: Mitogen-Activated Protein Kinases / Extracellular Signal-Regulated Kinases
16. Akt1/PKB Protein Kinase B
17. ROCK I. Rho-associated Coiled-coil containing Protein Kinase

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Receptor	Ligand	Adaptor	Target
Fas/CD95/APO-1	Fas-L/APO-1L	FADD/MORT1	Procaspase-8, apoptosis
TNF-R1	TNFα	TRADD+FADD	Procaspase-8, apoptosis
TNF-R1	TNFα	TRADD+RIP1+TRAF2	MEKK, Jun/Ap1, cell proliferation, IKK, NF-κB, inflammation, c-IAPs
TNF-R2/CD40	TNFα	TRAF2+TRAF1	MEKK, Jun/Ap1, cell proliferation, IKK, NF-κB, inflammation, c-IAPs
DR3/APO-3	APO-3L	FADD?	Procaspase-8, apoptosis
DR4	TRAIL/APO-2L	FADD	Procaspase-8, apoptosis
DR5	TRAIL/APO-2L	FADD	Procaspase-8, apoptosis
DcR1	TRAIL/APO-2L	none	decoy receptor, ligand sequestration
DcR2	TRAIL/APO-2L	none	decoy receptor, ligand sequestration
DcR3	Fas-L/APO-1L	none	decoy receptor, ligand sequestration

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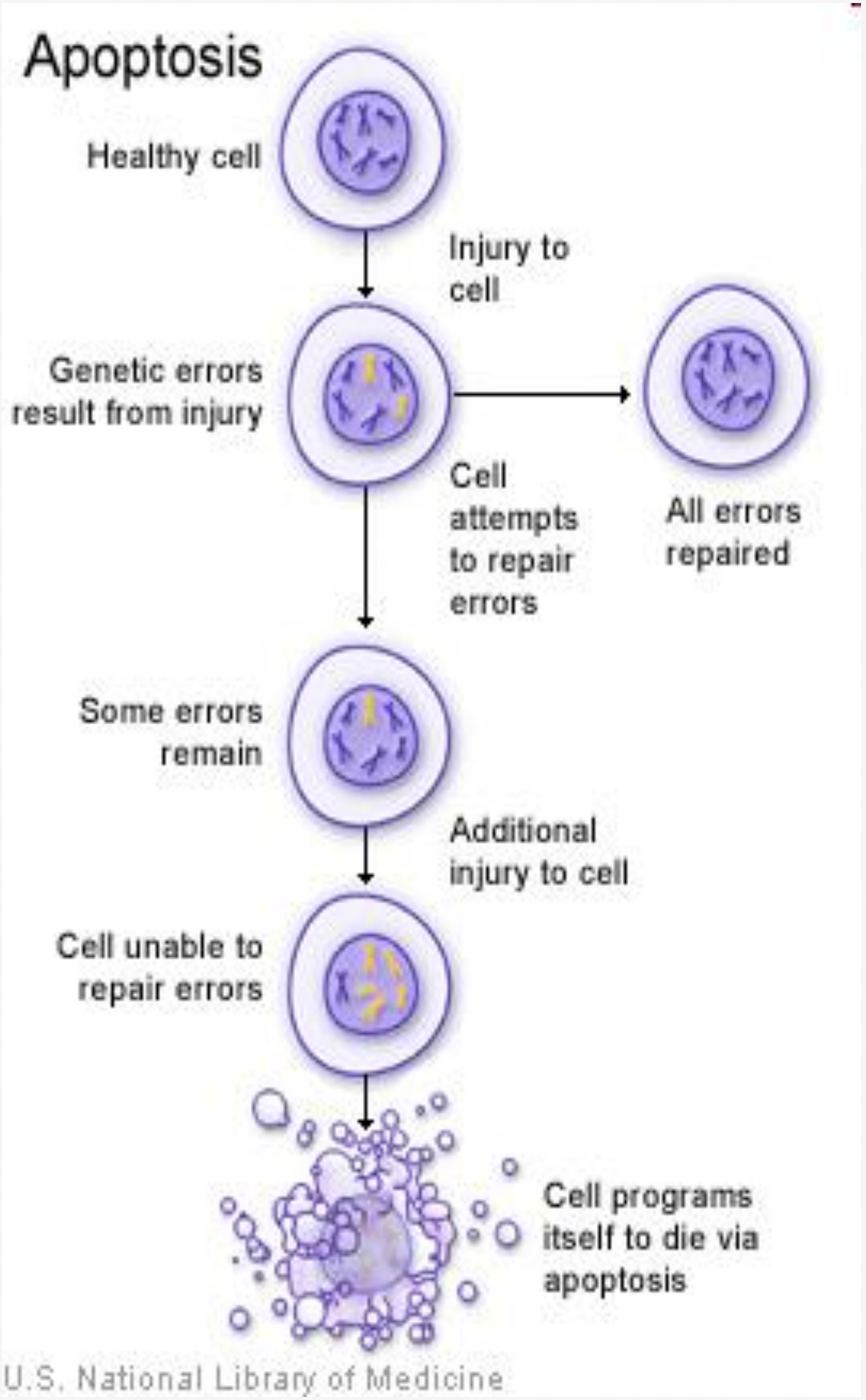
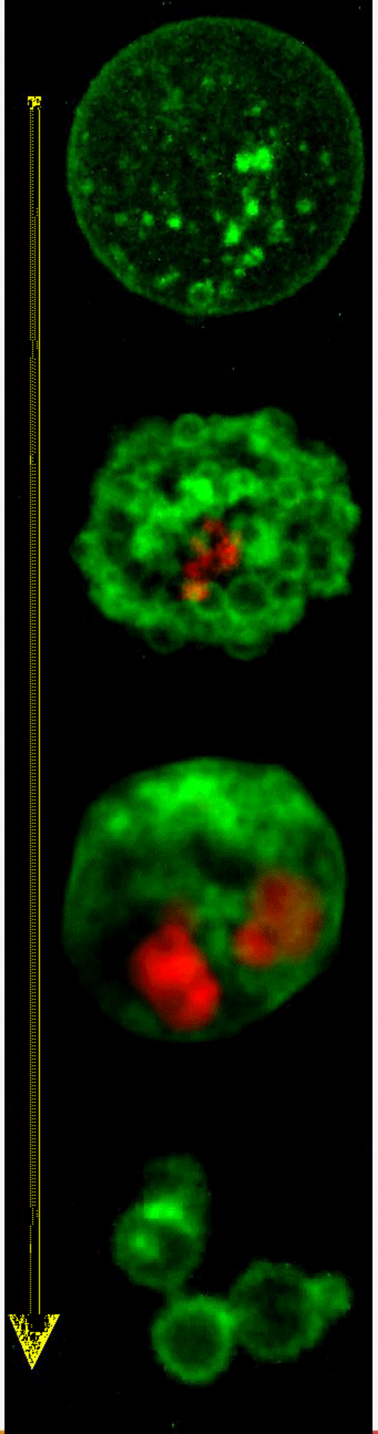
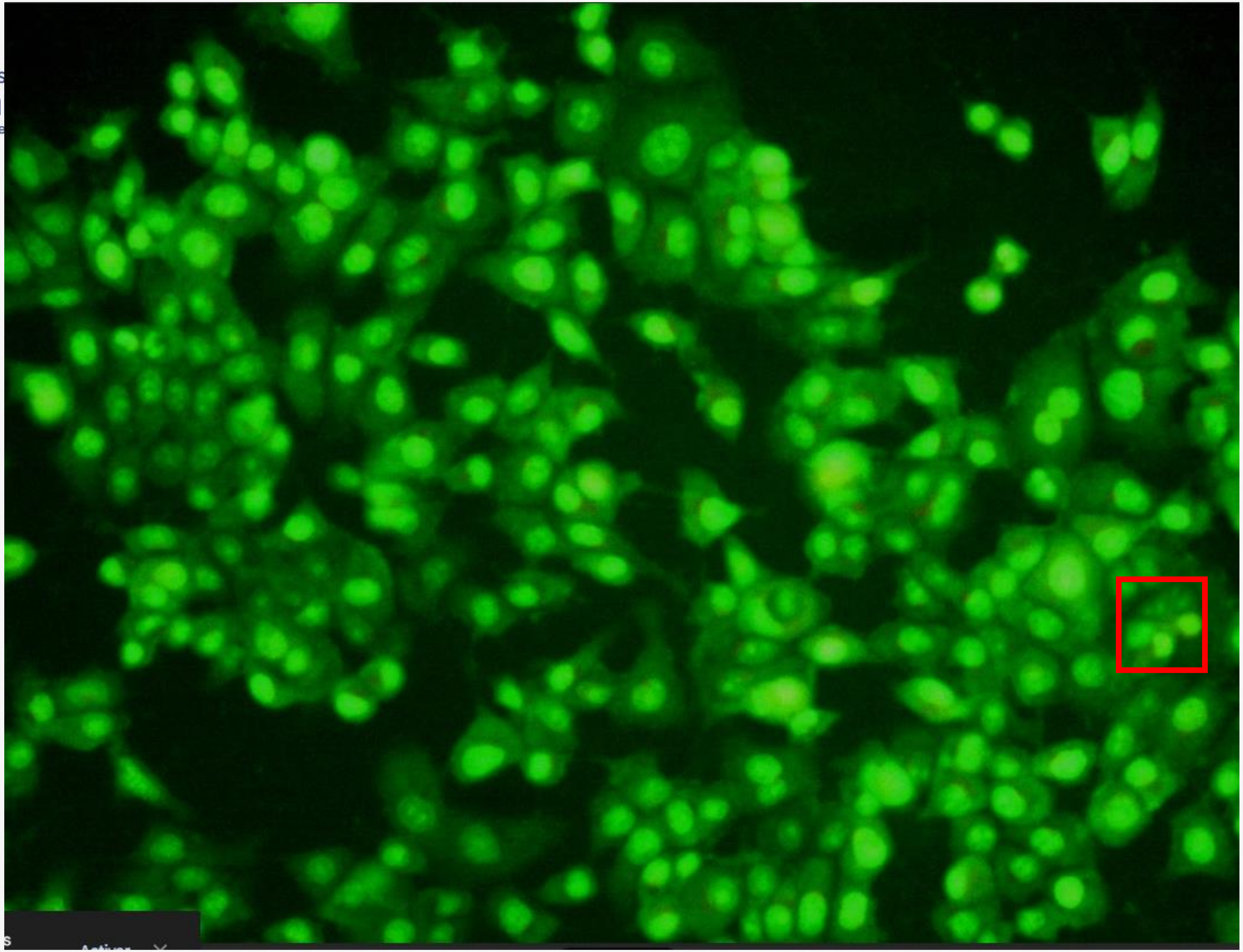
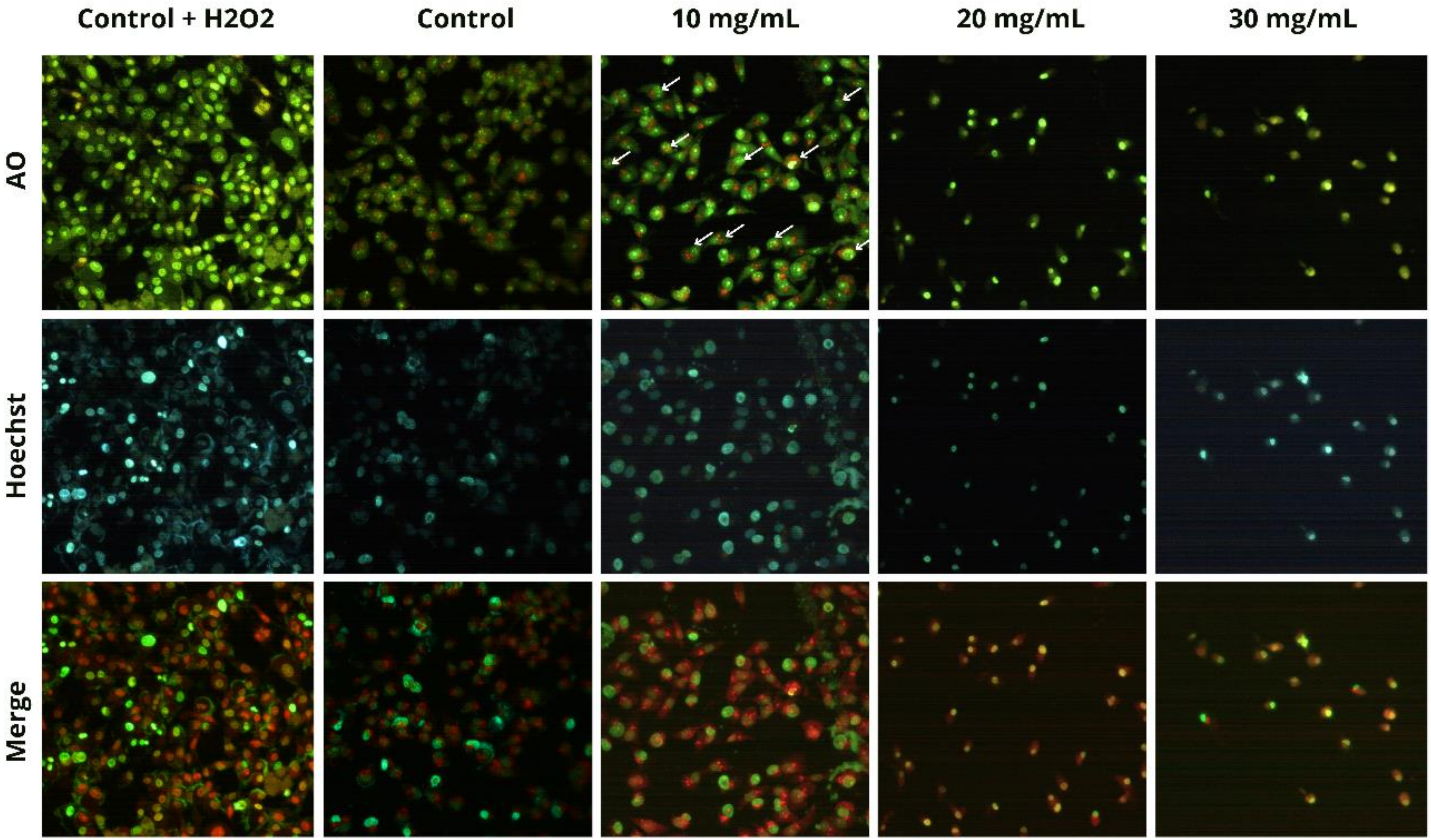
Si la célula no puede reparar los daños, entra en apoptosis








Control + H2O2
Control
10 mg/mL
20 mg/mL
30 mg/mL
AO
Hoechst
Merge

Proteome Profiler™ Array

Human Apoptosis Array Kit

Catalog Number ARY009

For the parallel determination of the relative levels of human apoptosis-related proteins.

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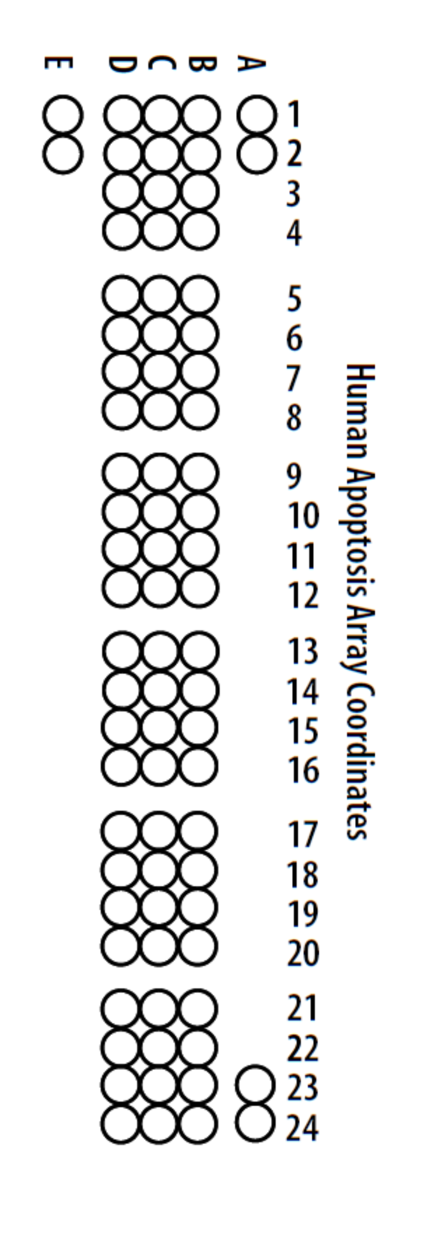
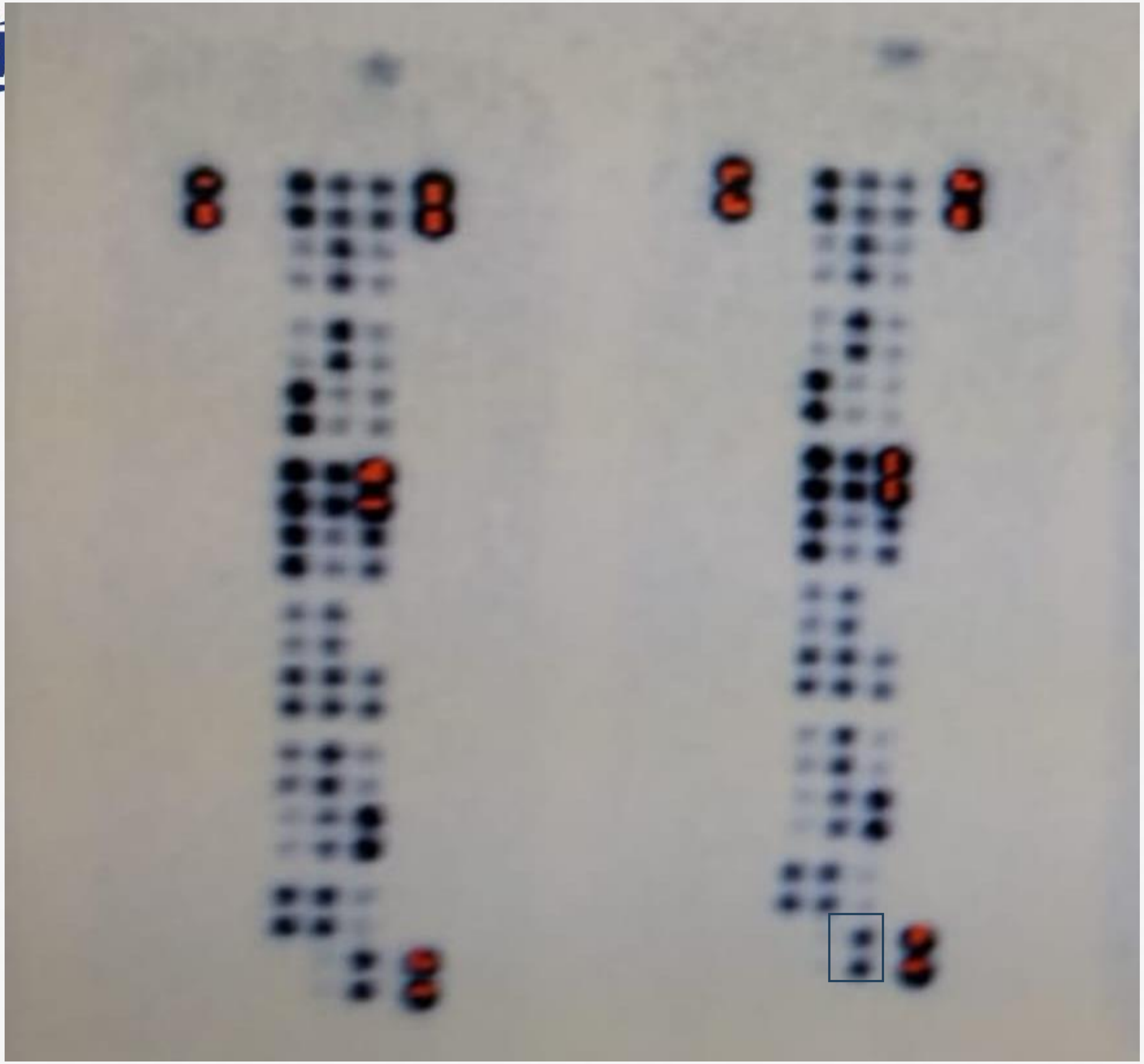
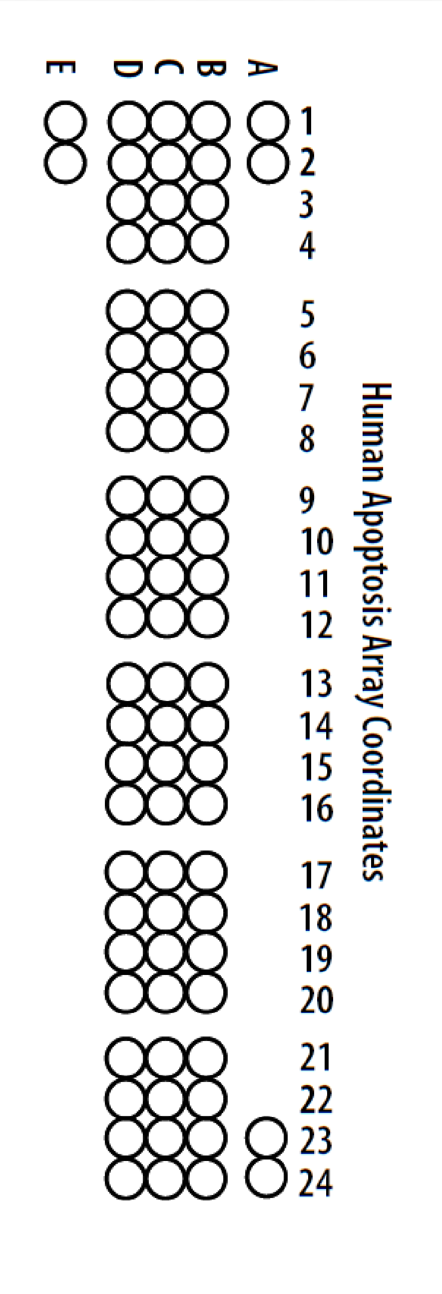
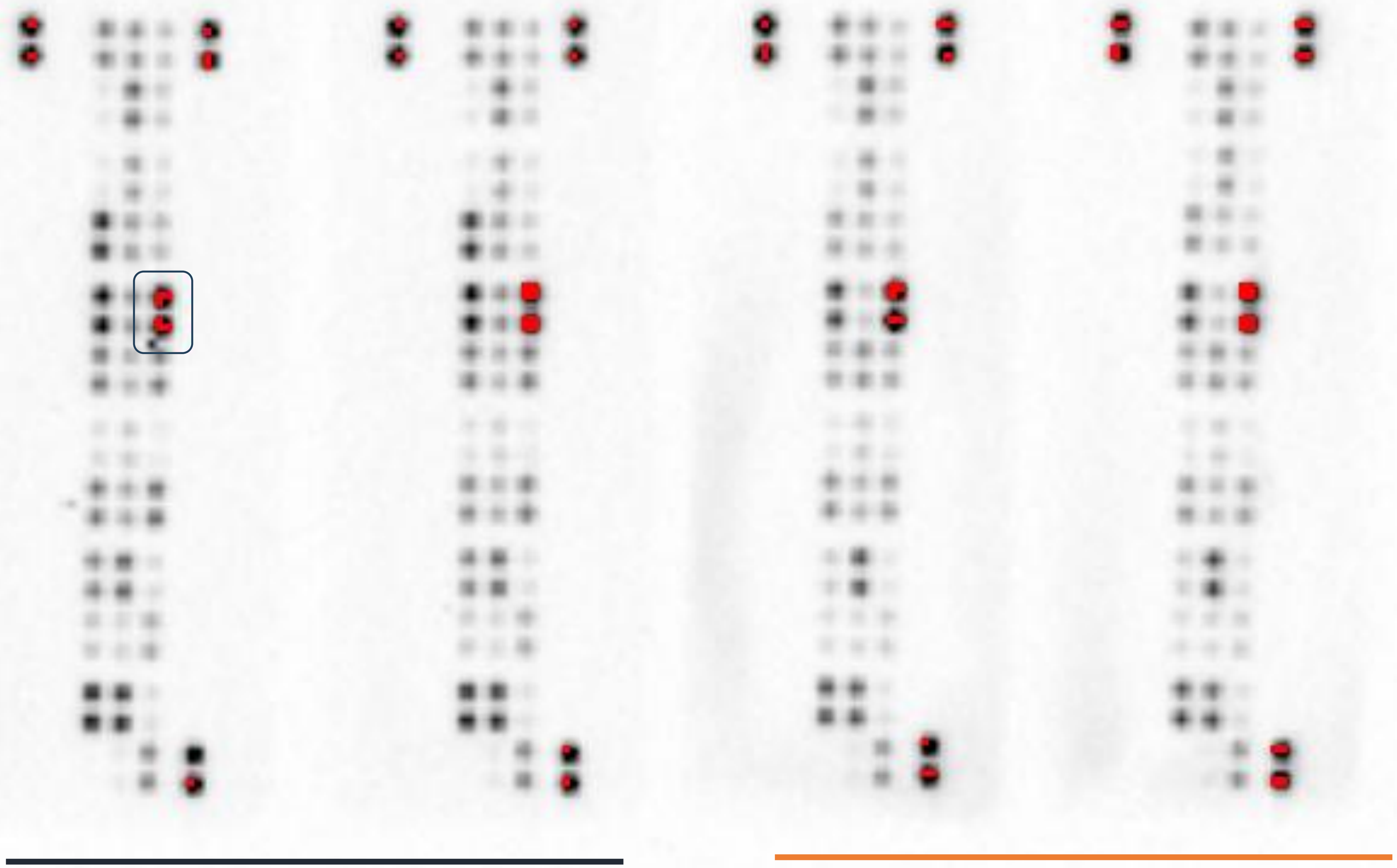
Proteome Profiler™ Array   Human Apoptosis Array Kit

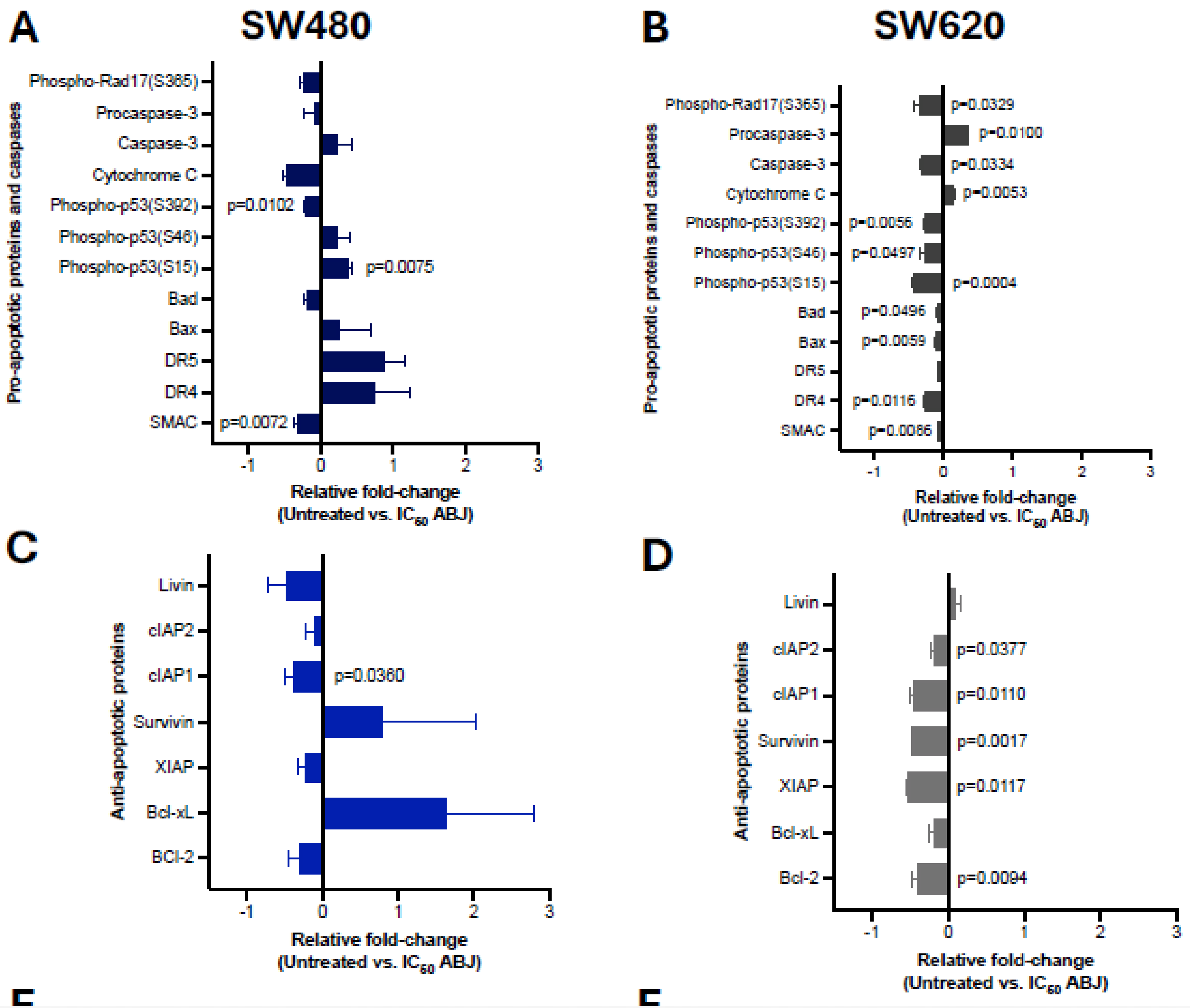


Coordinate	Target/Control	Coodinate	Target/Control
A1, A2	Reference Spots	C13, C14	HO-2/HMOX2
A23, A24	Reference Spots	C15, C16	HSP27
B1, B2	Bad	C17, C18	HSP60
B3, B4	Bax	C19, C20	HSP70
B5, B6	Bcl-2	C21, C22	HTRA2/Omi
B7, B8	Bcl-x	C23, C24	Livin
B9, B10	Pro-Caspase-3	D1, D2	PON2
B11, B12	Cleaved Caspase-3	D3, D4	p21/CIP1/CDKN1A
B13, B14	Catalase	D5, D6	p27/Kip1
B15, B16	cIAP-1	D7, D8	Phospho-p53 (S15)
B17, B18	cIAP-2	D9, D10	Phospho-p53 (S46)
B19, B20	Claspin	D11, D12	Phospho-p53 (S392)
B21, B22	Clusterin	D13, D14	Phospho-Rad17 (S635)
B23, B24	Cytochrome c	D15, D16	SMAC/Diablo
C1, C2	TRAIL R1/DR4	D17, D18	Survivin
C3, C4	TRAIL R2/DR5	D19, D20	TNF RI/TNFRSF1A
C5, C6	FADD	D21, D22	XIAP
C7, C8	Fas/TNFRSF6/CD95	D23, D24	PBS (Negative Control)
C9, C10	HIF-1α	E1, E2	Reference Spots
C11, C12	HO-1/HMOX1/HSP32		




Human Apoptosis Array Coordinates
A
B
C
D
E
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24




A
SW480
B
SW620
Phospho-Rad17(S365)
Procaspase-3
Caspase-3
Cytochrome C
Phospho-p53(S392)
Phospho-p53(S46)
Phospho-p53(S15)
Bad
Bax
DR5
DR4
SMAC
p=0.0102
p=0.0075
p=0.0072
Pro-apoptotic proteins and caspases
p=0.0329
p=0.0100
p=0.0334
p=0.0053
p=0.0056
p=0.0497
p=0.0004
p=0.0496
p=0.0059
p=0.0116
p=0.0086
-1
0
1
2
3
Relative fold-change
(Untreated vs. IC60 ABJ)
C
D
Livin
cIAP2
cIAP1
Survivin
XIAP
Bcl-xL
BCl-2
Bcl-2
p=0.0360
Anti-apoptotic proteins
p=0.0377
p=0.0110
p=0.0017
p=0.0117
p=0.0094

https://www.sciencedirect.com/science/article/pii/S0963996922003015

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